

End-to-End Student Admissions, Call Recording & Backend Architecture Handbook

A Comprehensive Blueprint: Public Marketing Ingestion • Telephony & AI Call Recording • Conversion & Portal Access • Automated Fee Installments • Database Schema

1. Executive Lifecycle Architecture Overview

The QuantNexa AI CRM automates the entire student journey from initial marketing engagement to active enrollment and fee collection. Below is the master end-to-end lifecycle flow:

Stage	Operational Action	Responsible Role	Backend Event / Trigger
1. Marketing Lead Capture	Student submits Public Form (Instagram Bio, Meta Ads, Google Search Ads, Landing Pages)	Prospect / Meta Ads / Webhook	POST /api/v1/leads/public-submit (UTM Captured)
2. Ingestion & AI Scoring	Lead ingested into CRM, AI Scores (0-100) Assigned, Disqualified or Created	System / AI Engine	Webhook Alert to Super Admin
3. Counselor Assignment	Automatic Round-Robin or Manual Assignment to Counselor	System / Counselor	Lead Owner Updated -> Assigned Counselor
4. Outbound Call & Recording	Counselor initiates Click-to-Call via Twilio or Exotel Cloud Telephony	Counselor / Twilio / Exotel	Call Initiated -> Dual Channel MP3 Recorded
5. Speech-to-Text & AI Call Analysis	Call recording processed by OpenAI Whisper & AI Engine	AI Backend / AI Engine	Call Transcript & Sentiment (Interested/Objective)
6. Student Conversion	Counselor clicks 'Convert to Student' on Counselor Interface	Counselor / System	Lead -> Student Record Created + Credentials Generated
7. Portal Access & Onboarding	Loading credentials & Welcome Pack, Automated Email App & WhatsApp API & SMTP Triggered (Student Portal Access)	System / Email App & WhatsApp	App API & SMTP Triggered (Student Portal Access)
8. Financials & Installment Fees	Fee structure set, Razorpay payment link generated, Counselor notified	System / Razorpay / Counselor	Payment Link Created -> Auto Fee Status Change

2. Public Marketing & External Enrollment Form Infrastructure

To capture prospective students from marketing channels (Instagram Bio, Meta Ads, Google Search Ads, Landing Pages), we provide a standalone, embeddable Public Enrollment Form.

A. Public Form Formats & URL Structure:

- **Shareable URL:** https://portal.quantnexa.com/enroll?source=instagram&campaign=summer_2026
- **Form Fields:** Full Name, Mobile Phone (with OTP verification), Email Address, Highest Qualification, Preferred Course, Preferred Branch/Campus, UTM Parameters.

B. Public Webhook Submission Endpoint (Node.js Express / Next.js):

```
POST /api/v1/leads/public-submit
Content-Type: application/json

{
  "name": "Rahul Sharma",
  "phone": "+919876543210",
  "email": "rahul.sharma@example.com",
  "courseId": "course_fullstack_01",
  "centerId": "center_delhi_01",
  "source": "instagram",
  "utm": { "campaign": "summer_2026", "medium": "cpc", "adset": "tech_graduates" }
}
```

3. Lead Ingestion, AI Scoring & Counselor Allocation Engine

A. AI Lead Quality Scoring Matrix (0–100 Score):

Upon arrival, every lead is evaluated by an automated scoring engine:

Parameter	Condition / Value	Score Addition
Phone Verification	OTP verified during form submission	+25 Points
High-Demand Course	Full-Stack Development / Data Science selected	+20 Points
Source Quality	Referral (+25), Meta Ads (+15), Cold Call (+5)	+15 to +25 Points
Profile Completeness	Provided both Phone, Email, Qualification & City	+15 Points
High Intent Signal	Requested immediate counselling session	+15 Points

B. Counselor Assignment Rules:

- **Round-Robin Allocation:** Leads are distributed evenly among available counselors assigned to that specific course/center.
- **Workload Balancing:** Counselors with over 25 active pending leads are bypassed to prevent lead drop-off.
- **Super Admin Override:** Super Admins can re-assign leads at any time directly from the Leads Table or Pipeline View.

4. Telephony Integration, Click-to-Call & AI Recording Engine

To track counselor responsiveness and record all outbound communications, the portal integrates with Cloud Telephony providers (Twilio / Exotel / Plivo).

A. How Click-to-Call Works (Step-by-Step Flow):

1. Counselor clicks the phone icon  next to a lead in the CRM.

- 2. CRM sends Webhook request to Twilio/Exotel API (POST /api/v1/telephony/call).
- 3. Telephony bridge first dials the Counselor's mobile phone / WebRTC browser softphone.
- 4. As soon as the Counselor picks up, the bridge dials the Student's mobile number and connects both parties.
- 5. The entire dual-channel audio conversation is recorded securely on cloud storage.

B. Webhook Call Status & Recording Lifecycle:

Event	Provider Webhook Payload	CRM Automated Response
Call Initiated	<code><code>call.started (CallSid, To, From)</code></code>	Logs call record in lead activity timeline (Status: 'In Progress')
Call Answered	<code><code>call.connected (Duration: 0s)</code></code>	Starts real-time call duration timer in counselor dashboard
Call Ended	<code><code>call.completed (Duration: 184s)</code></code>	Updates call log with total duration (e.g. 3m 04s).
Recording Ready	<code><code>recording.available (RecordingUrl)</code></code>	Downloads MP3 to private S3 bucket & triggers Whisper transcription

C. AI Speech-to-Text & Sentiment Analysis:

Once the MP3 recording is saved, an asynchronous background worker processes the audio:

- **OpenAI Whisper API:** Transcribes audio into verbatim text (Hindi + English code-switched support).
- **Sentiment & Interest Extractor:** Analyzes key terms to automatically mark the call status: *High Interest, Price Objection, Follow-up Needed, Not Interested.*

5. Student Conversion, Credential Provisioning & Portal Access

When a counselor confirms student interest and initial fee commitment, they click '**Convert Lead to Student**'.

A. Conversion Logic & Automated Triggers:

- 1. **Status Upgrade:** Lead status changes from 'Interested / Counselling' to Student Enrolled.
- 2. **Student Entity Created:** A permanent Student record is created with a unique Student ID (e.g. QN-2026-8942).
- 3. **Credential Generation:** Random temporary password generated and hashed via bcrypt.
- 4. **Automated Welcome Message:** WhatsApp API (Twilio/Meta) & SMTP Email send login credentials:

WhatsApp Welcome Message Template:

"Welcome to QuantNexa AI, Rahul! ■ Your student account has been activated.
Portal URL: <https://student.quantnexa.com>
Username: **rahul.sharma@example.com**
Temp Password: **qnX#9821**
Assigned Counselor: Priya Verma (+91-9812345678)"

6. Financial Management, Installment Scheduling & Payment Webhooks

A. Fee Calculator & Installment Structure:

The finance engine automatically computes course fees, scholarships, and installment schedules:

Component	Formula / Logic	Example (Full-Stack Course)
Base Course Fee	Standard Course Price	■75,000
Scholarship Discount	Merit / Entrance Exam Score Discount	- ■10,000 (13.3%)
Net Total Payable	Base Fee - Scholarship - Custom Discount	■65,000
Installment 1 (Token)	Due immediately upon enrollment	■15,000 (Paid via Razorpay)
Installment 2	Due 30 days after enrollment	■25,000 (Due 26-Sep-2026)
Installment 3	Due 60 days after enrollment	■25,000 (Due 26-Oct-2026)

B. Automated Payment Gateway Webhook (Razorpay / Cashfree):

When a student pays via the Razorpay payment link sent to WhatsApp, the gateway emits a webhook:

```
POST /api/v1/payments/razorpay-webhook
Header: X-Razorpay-Signature (HMAC SHA256 Verification)

{
  "event": "payment.captured",
  "payload": {
    "payment": {
      "entity": { "id": "pay_Kx9281aZ", "amount": 1500000, "status": "captured", "notes": {
"studentId": "QN-2026-8942", "installmentNo": 1 } }
    }
  }
}
```

- **Automated System Actions:** Recalculates paidAmount, updates pendingAmount, logs payment transaction, sends GST invoice PDF to student email.

7. Comprehensive Database Schema & Backend Architecture

Below is the PostgreSQL Prisma ERD schema required to power this entire workflow:

Prisma Database Models (schema.prisma):

```
model Lead {
  id String @id @default(uuid())
  name String
  phone String @unique
  email String
  source String // instagram, meta_ads, google_ads, website
  status LeadStatus @default(NEW)
  leadScore Int @default(50)
  assignedToId String?
  assignedTo User? @relation(fields: [assignedToId], references: [id])
  calls CallLog[]
  student Student?
  createdAt DateTime @default(now())
}

model CallLog {
  id String @id @default(uuid())
  leadId String
  lead Lead @relation(fields: [leadId], references: [id])
  callSid String @unique
  duration Int // in seconds
  recordingUrl String?
  transcript String? @db.Text
  sentiment String? // HIGH_INTENT, PRICE_OBJECTION, NOT_INTERESTED
  createdAt DateTime @default(now())
}

model Student {
  id String @id @default(uuid()) // QN-2026-XXXX
  leadId String @unique
  lead Lead @relation(fields: [leadId], references: [id])
  courseId String
  totalFee Float
  paidAmount Float @default(0)
  pendingAmount Float
  installments Installment[]
  createdAt DateTime @default(now())
}
```